Algebraic Geometry and Galois Theory

Lecture Notes from June 6, 2026

Based on `Note Jun 6, 2026 algebraic geometry 1.pdf`

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1 Guiding Theme

These notes organize the source around one main idea: algebraic geometry turns geometric objects into fields of functions, and those fields can then be studied by field-theoretic methods. In the examples below, projective curves such as rational curves and elliptic curves give finite extensions of simpler function fields. This is the point at which algebraic geometry meets Galois theory.

The route is:

$$\text{variety} \xrightarrow{\text{regular functions}} \text{coordinate ring} \xrightarrow{\text{fractions}} \text{function field} \xrightarrow{\text{comparison}} \text{field extension}.$$

The final example is an elliptic curve E whose function field is a quadratic extension of $\mathbb{C}(x)$.

The source is handwritten. The main mathematical content has been written in a textbook style, while unreadable arrows, labels, or diagrams are explicitly marked as unclear. Several computations below are included as standard background calculations so that the algebra can be checked line by line; these are labeled as standard computations rather than as newly recovered source content.

2 Learning Goals and Notation

Prerequisites

These notes assume basic field theory, vector spaces, polynomial rings, and quotient rings. Some standard background from algebraic geometry and Riemann surface theory is used as context, especially projective space, homogeneous polynomials, and the fact that compact Riemann surfaces can be realized as projective algebraic curves.

Learning goals

After reading these notes, a student should be able to:

- explain how a finite field extension is a vector space over the base field;
- carry out explicit basis, coordinate-ring, fraction-field, and invariant-field calculations;
- define affine and projective varieties, coordinate rings, and function fields;
- use homogeneous coordinates on projective space;
- compute the images of the quadratic and cubic Veronese maps;
- pass from a projective Weierstrass equation to its affine chart;
- describe why the elliptic curve $y^2 = x^3 + ax + b$ gives a quadratic extension of $\mathbb{C}(x)$ with Galois group $\mathbb{Z}/2\mathbb{Z}$.

Notation

Notation	Meaning
k	a base field
$\mathbb{C}, \mathbb{Q}, \mathbb{F}_q$	the complex numbers, rationals, and a finite field
$\mathbb{A}^n_k$	affine n-space over k
$\mathbb{P}^N_k$	projective N-space over k
$\mathrm{V}(I)$	zero locus of an ideal I
$k[X]$	coordinate ring of an affine variety X
$K(X)$	function field of an irreducible variety X
$[x_0 : \cdots : x_N]$	homogeneous coordinates
E	the elliptic curve in Weierstrass form
$F \subseteq L$	an extension of fields
$\mathrm{Gal}(L/F)$	Galois group of L/F

3 Field Extensions as Linear Algebra

We begin with the field-theoretic side of the story. A finite extension of fields is not just a larger field: it is also a finite-dimensional vector space over the smaller field. This is the linear-algebraic structure that later appears when a map of curves produces an extension of function fields.

Definition 3.1. Let $F \subseteq E$ be fields. The extension E/F is called *finite* if E is finite-dimensional as a vector space over F:

$$[E : F] := \dim_F E < \infty.$$

It is *finitely generated* if there exist $\alpha_1, \ldots, \alpha_n \in E$ such that

$$E = F(\alpha_1, \ldots, \alpha_n).$$

Remark 3.2. The source emphasizes that finite field extensions can be studied "via linear algebra." If $F \subseteq E$, then E is an F-vector space: addition is the addition of the field E, and scalar multiplication is

$$F \times E \longrightarrow E, \qquad (a, v) \longmapsto av.$$

The multiplication on E is compatible with this vector-space structure: it is F-bilinear as a map

$$E \times E \longrightarrow E, \qquad (u, v) \longmapsto uv.$$

Thus a finite extension is simultaneously a field and a finite-dimensional vector space over the smaller field.

Standard computation 3.3 (Two extensions as vector spaces)**.** The following computations are standard background examples of the slogan "field extensions can be studied by linear algebra."

First consider $\mathbb{C}/\mathbb{R}$. As a real vector space,

$$\mathbb{C} = \mathbb{R} \cdot 1 \oplus \mathbb{R} \cdot i, \qquad [\mathbb{C} : \mathbb{R}] = 2.$$

If $z = a + bi$, then its coordinate column in the ordered basis $(1, i)$ is

$$[z]_{(1,i)} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

Multiplication by a fixed element $u = p + qi$ is an $\mathbb{R}$-linear map

$$m_u \colon \mathbb{C} \longrightarrow \mathbb{C}, \qquad z \longmapsto uz.$$

Since

$$(p + qi)(a + bi) = (pa - qb) + (qa + pb)i,$$

the matrix of m_u in the basis $(1, i)$ is

$$[m_u]_{(1,i)} = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}.$$

Thus field multiplication becomes an explicit linear transformation.

Similarly, let $d \in \mathbb{Q}$ be squarefree and not a square. Then

$$\mathbb{Q}(\sqrt{d}) = \mathbb{Q} \cdot 1 \oplus \mathbb{Q} \cdot \sqrt{d}, \qquad [\mathbb{Q}(\sqrt{d}) : \mathbb{Q}] = 2.$$

For $u = a + b\sqrt{d}$, multiplication by u is determined by

$$(a + b\sqrt{d})(r + s\sqrt{d}) = (ar + bds) + (as + br)\sqrt{d}.$$

Hence, in the ordered basis $(1, \sqrt{d})$,

$$[m_u]_{(1,\sqrt{d})} = \begin{pmatrix} a & bd \\ b & a \end{pmatrix}.$$

Remark 3.4 (Section summary). A field extension E/F can be studied by linear algebra when E is finite-dimensional over F. Later, a geometric map of curves will produce such an extension at the level of function fields.

TODO: unclear from source, page 1. The handwritten operation diagram compares $(F, +_F, \cdot_F)$ and $(E, +_E, \cdot_E)$, but several arrows and labels are not legible enough to transcribe with confidence.

4 Varieties and Function Fields

The geometric objects in these notes are varieties: spaces cut out by polynomial equations. The algebra attached to such a space is its coordinate ring, and the field of fractions of that ring is its function field.

Definition 4.1. Let k be a field. An affine algebraic set in $\mathbb{A}^n_k$ is the common zero locus

$$\mathrm{V}(I) = \{p \in \mathbb{A}^n_k \mid f(p) = 0 \text{ for every } f \in I\}$$

of an ideal $I \subseteq k[x_1, \ldots, x_n]$. In these notes, an affine variety is an algebraic set whose vanishing ideal is prime. Equivalently, its coordinate ring is an integral domain.

Definition 4.2. A smooth n-manifold is a topological space M which is locally modeled on Euclidean space. More precisely, for every $p \in M$ there is an open neighborhood $U_p \subseteq M$ and a homeomorphism

$$U_p \cong \mathbb{R}^n$$

compatible with a differentiable atlas.

Remark 4.3. The contrast in the source is the following. A smooth manifold is locally modeled on $\mathbb{R}^n$, whereas an algebraic variety is locally or globally described by polynomial equations. The geometric intuition can look similar, but the definitions impose different kinds of structure.

Definition 4.4. Let $X = \mathrm{V}(I) \subseteq \mathbb{A}^n_k$ be an affine variety with I prime. Its coordinate ring is

$$k[X] = k[x_1, \ldots, x_n]/I.$$

Since I is prime, $k[X]$ is an integral domain. Its *function field* is

$$K(X) := \mathrm{Frac}(k[X]).$$

Remark 4.5. Elements of $K(X)$ are rational functions on X. They can be thought of as quotients f/g of polynomial functions, with g not identically zero. This is the algebraic-geometric analogue of passing from a space to the field of functions on it.

Example 4.6. For the affine line $X = \mathbb{A}^1_{\mathbb{C}}$, the coordinate ring is $\mathbb{C}[x]$, so

$$K(\mathbb{A}^1_{\mathbb{C}}) = \operatorname{Frac}(\mathbb{C}[x]) = \mathbb{C}(x).$$

Thus $\mathbb{C}(x)$ is the field of rational functions in one variable.

Standard computation 4.7 (A coordinate ring computed from a parametrized parabola)**.** This is a standard computation illustrating how an equation becomes a quotient ring. Let

$$X = \mathrm{V}(y - x^2) \subseteq \mathbb{A}^2_{\mathbb{C}}.$$

The coordinate ring is

$$\mathbb{C}[X] = \mathbb{C}[x, y]/(y - x^2).$$

Define a $\mathbb{C}$-algebra homomorphism

$$\Phi\colon \mathbb{C}[x, y] \longrightarrow \mathbb{C}[t], \qquad \Phi(x) = t, \quad \Phi(y) = t^2.$$

Then

$$\Phi(y - x^2) = t^2 - t^2 = 0,$$

so $(y - x^2) \subseteq \ker \Phi$. Conversely, every polynomial $f(x, y) \in \mathbb{C}[x, y]$ can be reduced modulo $y - x^2$ by replacing y with x^2. Thus

$$f(x, y) \equiv f(x, x^2) \pmod{(y - x^2)}.$$

If $\Phi(f) = f(t, t^2) = 0$ as a polynomial in t, then

$$f(x, x^2) = 0,$$

so the reduction of f is zero and $f \in (y - x^2)$. Hence

$$\ker \Phi = (y - x^2).$$

By the first isomorphism theorem,

$$\mathbb{C}[x, y]/(y - x^2) \cong \mathbb{C}[t].$$

Therefore

$$K(X) = \operatorname{Frac}(\mathbb{C}[X]) \cong \operatorname{Frac}(\mathbb{C}[t]) = \mathbb{C}(t).$$

In this example the coordinate t is the parameter, while the coordinate functions on X satisfy $x = t$ and $y = t^2$.

Standard computation 4.8 (A second affine example)**.** Let

$$U = \mathrm{V}(xy - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}.$$

The equation $xy = 1$ allows $y = x^{-1}$. Algebraically,

$$\mathbb{C}[U] = \mathbb{C}[x, y]/(xy - 1).$$

The homomorphism

$$\Psi\colon \mathbb{C}[x, y] \longrightarrow \mathbb{C}[t, t^{-1}], \qquad \Psi(x) = t, \quad \Psi(y) = t^{-1}$$

satisfies $\Psi(xy-1)=0$, and the same elimination argument gives

$$\mathbb{C}[x,y]/(xy-1) \cong \mathbb{C}[t,t^{-1}].$$

Its function field is

$$K(U) = \mathrm{Frac}(\mathbb{C}[t,t^{-1}]) = \mathbb{C}(t).$$

This example shows that different affine curves may have the same function field even though their coordinate rings are not the same.

Remark 4.9 (Section summary). The coordinate ring records polynomial functions on a variety. Passing to its fraction field allows quotients of polynomial functions and produces the function field.

TODO: unclear from source, page 2. The page contains illustrative local-chart drawings and a quotient-like picture, but the precise equivalence relation and labels in those drawings are not fully readable.

5 Projective Space and Projective Varieties

Projective space is introduced because many naturally occurring curves are best understood after adding their points at infinity. This is also the setting in which compact Riemann surfaces become algebraic curves: the embedding theorem places them inside complex projective space.

Definition 5.1. Let k be a field and $N \geq 0$. Projective N-space over k is

$$\mathbb{P}^N_k = \big(k^{N+1} \setminus \{(0,\dots,0)\}\big)/\!\sim,$$

where

$$(x_0,\dots,x_N) \sim (y_0,\dots,y_N)$$

if and only if there exists $\lambda \in k^\times$ such that

$$(y_0,\dots,y_N) = (\lambda x_0,\dots,\lambda x_N).$$

The equivalence class of a nonzero vector is written

$$[x_0 : \cdots : x_N].$$

Remark 5.2. The brackets $[x_0 : \cdots : x_N]$ are homogeneous coordinates. Multiplying all coordinates by the same nonzero scalar does not change the point:

$$[x_0 : \cdots : x_N] = [\lambda x_0 : \cdots : \lambda x_N], \qquad \lambda \in k^\times.$$

This is why equations on projective space must be homogeneous: if $f(x_0,\dots,x_N)=0$, then the truth of the equation should not depend on which representative of $[x_0 : \cdots : x_N]$ is chosen.

Standard computation 5.3 (Rescaling and homogeneous equations)**.** Let

$$F(X,Y,Z) = XZ - Y^2.$$

This polynomial is homogeneous of degree 2. If $[X:Y:Z] = [\lambda X : \lambda Y : \lambda Z]$, then

$$F(\lambda X, \lambda Y, \lambda Z) = (\lambda X)(\lambda Z) - (\lambda Y)^2 = \lambda^2(XZ - Y^2) = \lambda^2 F(X,Y,Z).$$

Since $\lambda \neq 0$, the equation $F = 0$ is true for one representative if and only if it is true for every representative. Therefore

$$XZ - Y^2 = 0$$

is a well-defined equation on $\mathbb{P}^2_{\mathbb{C}}$.

By contrast, the nonhomogeneous equation $X + 1 = 0$ is not well-defined on $\mathbb{P}^1_{\mathbb{C}}$. The two vectors $(-1, 1)$ and $(-2, 2)$ represent the same projective point:

$$[-1 : 1] = [-2 : 2].$$

But

$$-1 + 1 = 0, \qquad -2 + 1 = -1 \neq 0.$$

Thus the truth of the equation depends on the representative, so it cannot define a subset of projective space.

Definition 5.4. A projective algebraic set in $\mathbb{P}^N_k$ is the common zero locus of homogeneous polynomials. If $I \subseteq k[X_0, \ldots, X_N]$ is a homogeneous ideal, write

$$\mathrm{V}(I) = \{[x_0 : \cdots : x_N] \in \mathbb{P}^N_k \mid f(x_0, \ldots, x_N) = 0 \text{ for every } f \in I\}.$$

An irreducible projective algebraic set is a projective variety.

Remark 5.5 (Explanatory background). Projective space is covered by affine charts. For example, the chart $x_i \neq 0$ in $\mathbb{P}^N_k$ is obtained by scaling so that $x_i = 1$. This is the projective analogue of choosing coordinates on a local patch.

Example 5.6. The source lists several base fields that may be used in projective geometry:

$$\mathbb{C}, \qquad \mathbb{Q}, \qquad \mathbb{F}_q, \qquad \text{number fields.}$$

Remark 5.7 (Section summary). Projective geometry uses homogeneous coordinates and homogeneous equations. It is the natural setting for compact algebraic curves because it includes points at infinity.

6 Realizing Smooth Manifolds as Varieties

The source next asks when familiar smooth spaces can be described by polynomial equations. This is false for arbitrary smooth manifolds, but true in the one-dimensional complex case that matters here.

Remark 6.1. Not every smooth manifold can be realized as an algebraic variety. The source records compact Riemann surfaces as an important positive case.

Theorem 6.2 (Compact Riemann surfaces as projective curves)**.** *Let M be a compact Riemann surface. Then M can be embedded into complex projective space:*

$$\varphi \colon M \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$$

for some N. Its image is a closed complex submanifold of $\mathbb{P}^N_{\mathbb{C}}$, and by Chow's theorem it is a closed algebraic subvariety. Thus M can be regarded as a smooth projective algebraic curve over $\mathbb{C}$.

Remark 6.3. The conclusion is that, after choosing such an embedding,

$$M \cong \mathrm{im}(\varphi) \subseteq \mathbb{P}^N_{\mathbb{C}}.$$

This is the bridge from the analytic object M to an algebraic object defined by homogeneous equations.

Remark 6.4 (Section summary). Compact Riemann surfaces are special: although arbitrary smooth manifolds need not be algebraic, compact Riemann surfaces can be embedded as projective algebraic curves over $\mathbb{C}$.

TODO: unclear from source, page 3. The handwritten notes include an embedding diagram $M \to \mathbb{P}^N_{\mathbb{C}}$ and comments about injectivity/embedding. The main meaning is clear, but some side annotations are not readable enough to transcribe exactly.

7 The Projective Line

The simplest projective curve is $\mathbb{P}^1_{\mathbb{C}}$. It is the main comparison object for the later elliptic curve: maps to $\mathbb{P}^1_{\mathbb{C}}$ correspond to subfields isomorphic to $\mathbb{C}(x)$ inside a curve's function field.

Example 7.1. The complex projective line is

$$\mathbb{P}^1_{\mathbb{C}} = \big(\mathbb{C}^2 \setminus \{(0,0)\}\big)/\mathbb{C}^\times.$$

It is identified with the Riemann sphere:

$$\mathbb{P}^1_{\mathbb{C}} \cong S^2.$$

Remark 7.2 (Affine charts on $\mathbb{P}^1$). The two standard affine charts are obtained by choosing a nonzero coordinate. If $y \neq 0$, then

$$[x : y] = [x/y : 1].$$

If $x \neq 0$, then

$$[x : y] = [1 : y/x].$$

Thus $\mathbb{P}^1_{\mathbb{C}}$ is obtained by gluing two copies of the affine line.

Example 7.3. On the chart $y \neq 0$, the coordinate is $t = x/y$. On the chart $x \neq 0$, the coordinate is $s = y/x$. On the overlap both coordinates are nonzero and satisfy

$$s = 1/t.$$

Standard computation 7.4 (The function field of $\mathbb{P}^1_{\mathbb{C}}$)**.** This is a standard computation behind the later notation $K(\mathbb{P}^1_{\mathbb{C}}) = \mathbb{C}(x)$. On the affine chart $y \neq 0$, write

$$t = x/y.$$

Rational functions on this chart are elements of $\mathbb{C}(t)$. On the other chart $x \neq 0$, write

$$s = y/x.$$

On the overlap, $s = 1/t$, so

$$\mathbb{C}(s) = \mathbb{C}(1/t) = \mathbb{C}(t).$$

Thus the two affine charts give the same rational function field:

$$K(\mathbb{P}^1_{\mathbb{C}}) = \mathbb{C}(t).$$

Later the same field is written $\mathbb{C}(x)$ when the coordinate comes from the x-coordinate map on the elliptic curve.

Remark 7.5 (Section summary). The curve $\mathbb{P}^1_{\mathbb{C}}$ is the projective line, or Riemann sphere. Its function field is $\mathbb{C}(x)$, so it becomes the base curve for the later degree-two map from an elliptic curve.

8 Veronese Embeddings and Rational Curves

The Veronese maps are the first concrete examples of parametrized projective curves. They show how a curve that is abstractly $\mathbb{P}^1_{\mathbb{C}}$ can appear inside a higher-dimensional projective space as the zero set of homogeneous polynomial equations.

Example 8.1 (The quadratic Veronese embedding)**.** Define

$$\nu_2 \colon \mathbb{P}^1_{\mathbb{C}} \longrightarrow \mathbb{P}^2_{\mathbb{C}}, \qquad [x : y] \longmapsto [x^2 : xy : y^2].$$

This is well-defined because replacing (x, y) by $(\lambda x, \lambda y)$ multiplies all three target coordinates by the common scalar λ^2. Writing homogeneous coordinates on $\mathbb{P}^2_{\mathbb{C}}$ as $[X : Y : Z]$, every point in the image satisfies

$$XZ - Y^2 = 0.$$

Indeed,

$$(x^2)(y^2) - (xy)^2 = 0.$$

Proposition 8.2. *The map ν_2 is injective, and its image is the conic*

$$\mathrm{V}(XZ - Y^2) \subseteq \mathbb{P}^2_{\mathbb{C}}.$$

Proof. First suppose

$$\nu_2([x : y]) = \nu_2([x' : y']).$$

If $x \neq 0$, then the first and second coordinates of $[x^2 : xy : y^2]$ determine y/x. Hence $[x : y] = [1 : y/x]$ is determined by the image. If $x = 0$, then the image is $[0 : 0 : 1]$, so the preimage is $[0 : 1]$. Thus ν_2 is injective.

We have already checked that $\mathrm{im}(\nu_2) \subseteq \mathrm{V}(XZ - Y^2)$. Conversely, let $[X : Y : Z] \in \mathrm{V}(XZ - Y^2)$. If $X \neq 0$, then

$$[X : Y : Z] = [1 : Y/X : Z/X],$$

and the equation gives $Z/X = (Y/X)^2$. Hence

$$[X : Y : Z] = [1 : t : t^2] = \nu_2([1 : t]) \qquad \text{with } t = Y/X.$$

If $X = 0$, then the equation gives $Y = 0$, so the point is $[0 : 0 : 1] = \nu_2([0 : 1])$. Therefore the conic is exactly the image. □

Standard computation 8.3 (Recovering the parameter for the conic)**.** On the affine chart $x \neq 0$ of the source, every point of $\mathbb{P}^1_{\mathbb{C}}$ has the form $[1 : t]$. Substituting into ν_2 gives

$$\nu_2([1 : t]) = [1 : t : t^2].$$

Thus on the target chart $X \neq 0$, the parameter is recovered by

$$t = \frac{Y}{X}, \qquad \frac{Z}{X} = t^2.$$

The single source point with $x = 0$ is $[0 : 1]$, and its image is

$$\nu_2([0 : 1]) = [0 : 0 : 1].$$

This is the point at infinity on the conic relative to the affine chart $X \neq 0$. The computation shows explicitly how the parametrization and the equation $XZ - Y^2 = 0$ encode the same curve.

Example 8.4 (The cubic Veronese embedding)**.** Define

$$\nu_3 \colon \mathbb{P}^1_{\mathbb{C}} \longrightarrow \mathbb{P}^3_{\mathbb{C}}, \qquad [x : y] \longmapsto [x^3 : x^2y : xy^2 : y^3].$$

Writing homogeneous coordinates on $\mathbb{P}^3_{\mathbb{C}}$ as $[X : Y : Z : W]$, the source records the image as

$$\operatorname{im}(\nu_3) = \left\{ [X : Y : Z : W] \in \mathbb{P}^3_{\mathbb{C}} \;\middle|\; \begin{array}{l} Y^2 = XZ, \\ Z^2 = YW, \\ YZ = XW \end{array} \right\}.$$

The equations follow by substituting

$$X = x^3, \qquad Y = x^2y, \qquad Z = xy^2, \qquad W = y^3.$$

For example, $Y^2 = x^4y^2 = XZ$, $Z^2 = x^2y^4 = YW$, and $YZ = x^3y^3 = XW$.

Proposition 8.5. *The three equations displayed above cut out the image of ν_3.*

Proof. The substitution in the previous example proves that every point of $\operatorname{im}(\nu_3)$ satisfies the three equations. Conversely, let $[X : Y : Z : W]$ satisfy them.

If $X \neq 0$, scale the point so that $X = 1$. Put $t = Y/X$. The equation $Y^2 = XZ$ gives $Z/X = t^2$, and the equation $YZ = XW$ gives $W/X = t^3$. Hence the point is

$$[1 : t : t^2 : t^3] = \nu_3([1 : t]).$$

If $X = 0$, then $Y^2 = XZ = 0$, so $Y = 0$. Then $Z^2 = YW = 0$, so $Z = 0$. The point is therefore $[0 : 0 : 0 : 1] = \nu_3([0 : 1])$. Thus the common zero locus is exactly $\operatorname{im}(\nu_3)$. □

Standard computation 8.6 (Recovering the parameter for the twisted cubic)**.** On the source chart $x \neq 0$, write a point as $[1 : t]$. Then

$$\nu_3([1 : t]) = [1 : t : t^2 : t^3].$$

On the target chart $X \neq 0$, the parameter is recovered from

$$t = \frac{Y}{X}.$$

The defining equations then force the remaining coordinates:

$$\frac{Z}{X} = \left(\frac{Y}{X}\right)^2 = t^2, \qquad \frac{W}{X} = \frac{YZ}{X^2} = t^3.$$

At the source point $[0 : 1]$, the image is

$$\nu_3([0 : 1]) = [0 : 0 : 0 : 1].$$

This is the point at infinity of the twisted cubic with respect to the chart $X \neq 0$.

Remark 8.7. The curves $\operatorname{im}(\nu_2)$ and $\operatorname{im}(\nu_3)$ are rational curves because they are parametrized by $\mathbb{P}^1_{\mathbb{C}}$. The quadratic case gives a conic in $\mathbb{P}^2_{\mathbb{C}}$, while the cubic case gives a twisted cubic in $\mathbb{P}^3_{\mathbb{C}}$.

Remark 8.8 (Section summary)*.* The Veronese maps show how a parametrized curve can be translated into implicit homogeneous equations. This prepares the way for studying curves by their equations rather than by pictures alone.

9 A Plane Cubic Model of an Elliptic Curve

The rational curves above have function field $\mathbb{C}(t)$. The next example is different: a nonsingular plane cubic is a genus-one curve. Analytically it is related to a complex torus, while algebraically it is described by a Weierstrass equation.

Example 9.1. The source records a map from a torus T into $\mathbb{P}^2_{\mathbb{C}}$ whose image is given by the homogeneous cubic equation

$$Y^2Z - X^3 - aXZ^2 - bZ^3 = 0.$$

Equivalently,

$$Y^2Z = X^3 + aXZ^2 + bZ^3.$$

The corresponding projective curve is

$$E = \mathrm{V}(Y^2Z - X^3 - aXZ^2 - bZ^3) \subseteq \mathbb{P}^2_{\mathbb{C}}.$$

When this cubic is nonsingular, it is an elliptic curve.

Remark 9.2. For the Weierstrass cubic above, nonsingularity is equivalent to the cubic polynomial

$$x^3 + ax + b$$

having distinct roots. Equivalently, its discriminant is nonzero:

$$-16(4a^3 + 27b^2) \neq 0.$$

This condition is not written explicitly in the source, but it is the standard condition needed for the phrase "elliptic curve" in this model.

Remark 9.3 (The affine chart $Z \neq 0$). On the inhomogeneous open set $Z \neq 0$, choose the representative with $Z = 1$. Put

$$x = X/Z, \qquad y = Y/Z.$$

Then the projective equation becomes the affine Weierstrass equation

$$y^2 = x^3 + ax + b.$$

Thus the affine part of E is

$$E \cap \{Z \neq 0\} = \{(x, y) \in \mathbb{C}^2 \mid y^2 = x^3 + ax + b\}.$$

Standard computation 9.4 (Dehomogenizing the cubic)**.** On the chart $Z \neq 0$, divide

$$Y^2Z = X^3 + aXZ^2 + bZ^3$$

by Z^3. This gives

$$\left(\frac{Y}{Z}\right)^2 = \left(\frac{X}{Z}\right)^3 + a\left(\frac{X}{Z}\right) + b.$$

With

$$x = X/Z, \qquad y = Y/Z,$$

the equation becomes

$$y^2 = x^3 + ax + b.$$

Conversely, any affine solution (x, y) gives the projective point

$$[X : Y : Z] = [x : y : 1],$$

and direct substitution gives

$$y^2 \cdot 1 = x^3 + ax \cdot 1^2 + b \cdot 1^3.$$

Thus the affine equation and the $Z \neq 0$ part of the projective equation are exactly the same calculation in different coordinates.

Remark 9.5 (The point at infinity). The points of E with $Z = 0$ satisfy

$$-X^3 = 0,$$

so $X = 0$. Since $[0 : 0 : 0]$ is not a point of projective space, $Y \neq 0$, and there is exactly one point at infinity:

$$[0 : 1 : 0].$$

Therefore

$$E = \{(x, y) \in \mathbb{C}^2 \mid y^2 = x^3 + ax + b\} \cup \{[0 : 1 : 0]\}.$$

Standard computation 9.6 (The point at infinity and the sign involution)**.** Set $Z = 0$ in the homogeneous equation:

$$Y^2Z - X^3 - aXZ^2 - bZ^3 = 0.$$

Every term containing Z vanishes, leaving

$$-X^3 = 0.$$

Hence $X = 0$. A projective point cannot have all coordinates zero, so $Y \neq 0$. Rescaling by Y^{-1} gives the unique point

$$[0 : Y : 0] = [0 : 1 : 0].$$

On the affine chart, the map

$$\iota(x, y) = (x, -y)$$

preserves the equation because

$$(-y)^2 = y^2 = x^3 + ax + b.$$

The point at infinity is also fixed by the corresponding projective map

$$[X : Y : Z] \longmapsto [X : -Y : Z],$$

since

$$[0 : -1 : 0] = [0 : 1 : 0].$$

Remark 9.7 (Section summary). The projective cubic contains the affine Weierstrass curve together with one point at infinity. When nonsingular, this projective curve is an elliptic curve.

Remark 9.8 (The 180-degree quotient picture). The source also draws a torus with a 180° rotation and a folding picture. This should be read as a quotient picture, not as a deformation that turns a torus itself into a sphere. Topologically, a complex elliptic curve is a torus. The order-two symmetry may be visualized as a 180° rotation identifying paired points. In the Weierstrass model, the same symmetry is the involution

$$\iota \colon E \longrightarrow E, \qquad \iota(x, y) = (x, -y).$$

Taking the quotient means that a point and its partner under ι are regarded as the same point. The quotient by this involution is the projective line $\mathbb{P}^1_{\mathbb{C}}$, which is the Riemann sphere. Algebraically, the quotient map is the two-to-one map given by the x-coordinate, developed in the next section.

TODO: unclear from source, page 6. The source draws a map from a torus T to the plane cubic, but it does not specify the formula for the map. These notes therefore record only the image equation. The 180° quotient picture is included above, but the source does not provide its full branch or fixed-point data.

10 The Function Field of the Elliptic Curve

The field-theoretic content of the elliptic-curve example appears when we compare the function field of E with the function field of $\mathbb{P}^1_{\mathbb{C}}$. The x-coordinate gives a two-to-one map $E \to \mathbb{P}^1_{\mathbb{C}}$, and this becomes a degree-two extension of function fields.

Definition 10.1. Let

$$E = \mathrm{V}(Y^2Z - X^3 - aXZ^2 - bZ^3) \subseteq \mathbb{P}^2_{\mathbb{C}}$$

be the nonsingular projective cubic above. As standard background, passing from an irreducible curve to a nonempty affine open subset does not change its function field. On the affine chart $Z \neq 0$, the coordinate ring is

$$\mathbb{C}[x, y]/(y^2 - x^3 - ax - b).$$

Its function field is the fraction field

$$K(E) = \mathrm{Frac}\big(\mathbb{C}[x, y]/(y^2 - x^3 - ax - b)\big).$$

Equivalently,

$$K(E) \cong \mathbb{C}(x)[T]/(T^2 - x^3 - ax - b),$$

where the class of T is the function y.

Standard computation 10.2 (Deriving $K(E) = \mathbb{C}(x)(y)$)**.** Let

$$f(x) = x^3 + ax + b.$$

On the affine chart $Z \neq 0$, the coordinate ring is

$$A = \mathbb{C}[x, y]/(y^2 - f(x)).$$

No nonzero polynomial in $\mathbb{C}[x]$ lies in the ideal $(y^2 - f(x))$, so the class of x gives a copy of $\mathbb{C}[x]$ inside A. In particular, inside the fraction field of A, the element x is transcendental over $\mathbb{C}$ and y satisfies

$$y^2 = f(x).$$

After passing from $\mathbb{C}[x]$ to its fraction field $\mathbb{C}(x)$, we obtain the localized algebra

$$\mathbb{C}(x)[y]/(y^2 - f(x)).$$

Every polynomial in y can be reduced to degree at most one using $y^2 = f(x)$. Explicitly,

$$y^{2m} = f(x)^m, \qquad y^{2m+1} = f(x)^m y.$$

Therefore every element of $\mathbb{C}(x)[y]/(y^2 - f(x))$ has the form

$$a(x) + b(x)y, \qquad a(x), b(x) \in \mathbb{C}(x).$$

Since $f(x)$ has distinct roots, it is not a square in $\mathbb{C}(x)$. Hence $T^2 - f(x)$ is irreducible over $\mathbb{C}(x)$, so this localized algebra is a field. It is therefore the fraction field of A, and

$$K(E) = \mathbb{C}(x)(y), \qquad y^2 = f(x).$$

Thus $K(E)$ is generated over $\mathbb{C}(x)$ by one square root, and every element of $K(E)$ can be written uniquely as $a(x) + b(x)y$.

Remark 10.3. The function field of the projective line is

$$K(\mathbb{P}^1_{\mathbb{C}}) = \mathbb{C}(x).$$

The rational function $x = X/Z$ on E has a pole at the point at infinity. By the standard fact that a nonconstant rational function on a nonsingular projective curve defines a morphism to $\mathbb{P}^1$, it gives a morphism

$$E \longrightarrow \mathbb{P}^1_{\mathbb{C}}.$$

On affine points this morphism is simply

$$(x, y) \longmapsto [x : 1],$$

and the point at infinity $[0 : 1 : 0]$ maps to $[1 : 0]$. At the level of function fields, it corresponds to the inclusion

$$\mathbb{C}(x) \subseteq K(E).$$

This is an example of the general reversal: a map of varieties gives an inclusion of function fields in the opposite direction.

$$E \xrightarrow{\quad \pi = x \quad} E/\langle \iota \rangle \cong \mathbb{P}^1_{\mathbb{C}}$$

$$K(E) \hookleftarrow K(E)^{\langle \iota^* \rangle} = K(\mathbb{P}^1_{\mathbb{C}}) = \mathbb{C}(x)$$

Remark 10.4 (The degree-two map). For a generic value of x, the equation

$$y^2 = x^3 + ax + b$$

has two solutions:

$$y = \pm\sqrt{x^3 + ax + b}.$$

Therefore the morphism $E \to \mathbb{P}^1_{\mathbb{C}}$ defined by x is a degree-two map. The source records this as a "2 : 1 map."

Proposition 10.5. *Let*

$$F = \mathbb{C}(x), \qquad L = K(E).$$

Then

$$L = F(y), \qquad y^2 = x^3 + ax + b,$$

and L/F is a quadratic extension. Moreover,

$$\mathrm{Gal}(L/F) \cong \mathbb{Z}/2\mathbb{Z}.$$

Proof. Since $L \cong \mathbb{C}(x)[T]/(T^2 - x^3 - ax - b)$, the field L is obtained from $F = \mathbb{C}(x)$ by adjoining a root y of

$$T^2 - (x^3 + ax + b) \in F[T].$$

Because $x^3 + ax + b$ has distinct roots, it has simple zeros as a rational function of x. A square in $\mathbb{C}(x)$ has every zero and pole of even order, so $x^3 + ax + b$ is not a square in F. Therefore

$$[L : F] = 2.$$

The nontrivial automorphism over F fixes x and sends

$$y \longmapsto -y.$$

Indeed, this assignment preserves the defining relation because

$$(-y)^2 = y^2 = x^3 + ax + b.$$

It has order two, and a quadratic extension has at most two F-automorphisms. Thus

$$\mathrm{Gal}(L/F) = \{1, \sigma\}, \qquad \sigma(x) = x, \quad \sigma(y) = -y,$$

so

$$\mathrm{Gal}(L/F) \cong \mathbb{Z}/2\mathbb{Z}.$$

□

Remark 10.6. Geometrically, the involution $\iota(x, y) = (x, -y)$ exchanges the two points in a generic fiber of the map $E \to \mathbb{P}^1_{\mathbb{C}}$. Algebraically, the induced automorphism ι^* is the nontrivial automorphism of the quadratic function-field extension $K(E)/\mathbb{C}(x)$.

Standard computation 10.7 (Computing the invariant field)**.** Write a general element of $K(E)$ over $\mathbb{C}(x)$ as

$$u = a(x) + b(x)y, \qquad a(x), b(x) \in \mathbb{C}(x).$$

The pullback action of the involution is

$$\iota^*(x) = x, \qquad \iota^*(y) = -y.$$

Therefore

$$\iota^*(u) = \iota^*\big(a(x) + b(x)y\big) = a(x) - b(x)y.$$

The element u is invariant precisely when

$$a(x) + b(x)y = a(x) - b(x)y.$$

Subtracting gives

$$2b(x)y = 0.$$

Since the ground field is $\mathbb{C}$, the characteristic is zero, and $y \neq 0$ in the function field. Hence $b(x) = 0$. Therefore the invariant elements are exactly

$$u = a(x) \in \mathbb{C}(x),$$

and so

$$K(E)^{\langle \iota^* \rangle} = \mathbb{C}(x).$$

Remark 10.8 (Invariant functions). In standard algebraic language, checking which quantities are unchanged by the 180° symmetry means taking invariants. The induced action on rational functions is

$$\iota^*\colon K(E) \longrightarrow K(E), \qquad \iota^*(x) = x, \quad \iota^*(y) = -y.$$

Then a rational function $f \in K(E)$ is unchanged by the symmetry precisely when

$$\iota^*(f) = f.$$

The fixed field of this order-two action is

$$K(E)^{\langle \iota^* \rangle} = \mathbb{C}(x).$$

An invariant rational function has the same value at paired points

$$P \quad \text{and} \quad \iota(P)$$

wherever both values are defined, so it descends to a rational function on the quotient curve. Thus the invariant functions are exactly the rational functions of the x-coordinate, which is why the quotient curve has function field $\mathbb{C}(x) = K(\mathbb{P}^1_{\mathbb{C}})$.

Remark 10.9 (Section summary). The elliptic curve function field is obtained from $\mathbb{C}(x)$ by adjoining one square root. The sign change of that square root is the nontrivial Galois automorphism, and the functions fixed by this sign change are exactly $\mathbb{C}(x)$.

TODO: unclear from source, page 8. The page indicates a two-to-one map and sketches the square root description of the fibers. Branch points or ramification data are not specified in the source.

11 Summary: From Geometry to Galois Theory

The lecture's main chain of ideas can now be read in one line:

$$E:\ y^2 = x^3 + ax + b \quad \rightsquigarrow \quad K(E) = \mathbb{C}(x)(y) \quad \rightsquigarrow \quad K(E)/\mathbb{C}(x).$$

The map $E \to \mathbb{P}^1_{\mathbb{C}}$ defined by the x-coordinate is generically two-to-one, because a typical value of x gives the two points

$$(x, \sqrt{x^3 + ax + b}) \qquad \text{and} \qquad (x, -\sqrt{x^3 + ax + b}).$$

On function fields, this means that $K(E)$ is obtained from $\mathbb{C}(x)$ by adjoining one square root:

$$y^2 = x^3 + ax + b.$$

The nontrivial symmetry of the extension fixes x and sends y to $-y$. Therefore the Galois group is the two-element group

$$\mathrm{Gal}(K(E)/\mathbb{C}(x)) \cong \mathbb{Z}/2\mathbb{Z}.$$

Equivalently, the base field $\mathbb{C}(x)$ is the field of rational functions on the quotient $E/\langle \iota \rangle \cong \mathbb{P}^1_{\mathbb{C}}$.

12 Exercises

Exercise 12.1. *Let $F \subseteq E$ be a finite field extension. Explain explicitly why E is an F-vector space, and identify the scalar multiplication map.*

Exercise 12.2. *For the extension $\mathbb{Q}(\sqrt{d})/\mathbb{Q}$, with d squarefree and not a square, write the matrix for multiplication by $a + b\sqrt{d}$ in the basis $(1, \sqrt{d})$.*

Exercise 12.3. *Compute the coordinate ring and function field of the affine parabola*

$$X = \mathrm{V}(y - x^2) \subseteq \mathbb{A}^2_{\mathbb{C}}.$$

Write the homomorphism to $\mathbb{C}[t]$, compute its kernel, and identify the fraction field.

Exercise 12.4. *Compute the coordinate ring and function field of*

$$U = \mathrm{V}(xy - 1) \subseteq \mathbb{A}^2_{\mathbb{C}}.$$

Explain why y becomes x^{-1} in the coordinate ring.

Exercise 12.5. *Show that the equation $XZ - Y^2 = 0$ is well-defined on $\mathbb{P}^2_{\mathbb{C}}$; that is, its truth is unchanged when $[X : Y : Z]$ is rescaled.*

Exercise 12.6. *Give an explicit example showing that the equation $X + 1 = 0$ is not well-defined on $\mathbb{P}^1_{\mathbb{C}}$.*

Exercise 12.7. *For the quadratic Veronese map*

$$\nu_2([x : y]) = [x^2 : xy : y^2],$$

verify by substitution that every image point satisfies $XZ - Y^2 = 0$. Then recover the parameter t on the chart $X \neq 0$.

Exercise 12.8. *Repeat the injectivity argument for the cubic Veronese map*

$$\nu_3([x : y]) = [x^3 : x^2y : xy^2 : y^3].$$

Exercise 12.9. *Verify directly that the three equations*

$$Y^2 = XZ, \qquad Z^2 = YW, \qquad YZ = XW$$

hold for every point in the image of ν_3.

Exercise 12.10. *On the chart $Z \neq 0$, check that the homogeneous equation*

$$Y^2Z = X^3 + aXZ^2 + bZ^3$$

reduces to $y^2 = x^3 + ax + b$. Then check separately that the only point of the curve with $Z = 0$ is $[0 : 1 : 0]$.

Exercise 12.11. *Verify directly that the transformation*

$$(x, y) \longmapsto (x, -y)$$

preserves the affine equation $y^2 = x^3 + ax + b$.

Exercise 12.12. *Let $f(x) = x^3 + ax + b$. In the field $\mathbb{C}(x)(y)$ with $y^2 = f(x)$, reduce*

$$c_0(x) + c_1(x)y + c_2(x)y^2 + c_3(x)y^3$$

to the form $A(x) + B(x)y$, where $A(x), B(x) \in \mathbb{C}(x)$.

Exercise 12.13. *Let ι^* be the automorphism of $K(E)$ determined by $\iota^*(x) = x$ and $\iota^*(y) = -y$. Using*

$$K(E) = \mathbb{C}(x) \oplus \mathbb{C}(x)y$$

as a vector space over $\mathbb{C}(x)$, show that the invariant subfield is

$$K(E)^{\langle \iota^* \rangle} = \mathbb{C}(x).$$

Exercise 12.14. *Let $L = \mathbb{C}(x)(y)$, where $y^2 = x^3 + ax + b$, and assume $x^3 + ax + b$ has distinct roots. Verify that the map*

$$\sigma(x) = x, \qquad \sigma(y) = -y$$

defines a nontrivial $\mathbb{C}(x)$-automorphism of L, and compute σ^2.

Exercise 12.15. *Explain in words why invariant rational functions are constant on paired points P and $\iota(P)$, and why this means they descend to rational functions on the quotient $E/\langle \iota \rangle$.*

Hints

1. Use the field operations of E: addition is addition in E, and scalar multiplication is $(a, v) \mapsto av$ for $a \in F$, $v \in E$.
2. Compute $(a + b\sqrt{d})(r + s\sqrt{d}) = (ar + bds) + (as + br)\sqrt{d}$.
3. Use the map $x \mapsto t$, $y \mapsto t^2$. The relation $y = x^2$ lets you eliminate y, so the coordinate ring is isomorphic to $\mathbb{C}[t]$.
4. Use the map $x \mapsto t$, $y \mapsto t^{-1}$. The relation $xy = 1$ makes y the inverse of x.
5. Substitute $(\lambda X, \lambda Y, \lambda Z)$ into $XZ - Y^2$.
6. Compare the two representatives $[-1 : 1]$ and $[-2 : 2]$.
7. Substitute $X = x^2$, $Y = xy$, and $Z = y^2$. On $X \neq 0$, use $t = Y/X$.
8. Work on the charts $x \neq 0$ and $x = 0$, as in the proof for ν_2.
9. Substitute $X = x^3$, $Y = x^2y$, $Z = xy^2$, and $W = y^3$.
10. Set $Z = 1$ on the affine chart. For $Z = 0$, the homogeneous equation forces $X = 0$, leaving the single projective point $[0 : 1 : 0]$.
11. The square of $-y$ is the same as the square of y.
12. Replace y^2 by $f(x)$ and y^3 by $f(x)y$.
13. Write a general element as $a(x) + b(x)y$. Applying ι^* gives $a(x) - b(x)y$, so invariance forces $b(x) = 0$.
14. Check that σ preserves the relation $y^2 = x^3 + ax + b$, then apply σ twice.
15. If $f(P) = f(\iota(P))$, then the value depends only on the equivalence class of P in the quotient.

Further Reading

The references below are standard background sources and are not part of the handwritten lecture source.

- Robin Hartshorne, *Algebraic Geometry*, Chapters I.1–I.3, for varieties, projective space, and morphisms.
- Igor Shafarevich, *Basic Algebraic Geometry 1*, for an introductory treatment of affine and projective varieties.
- William Fulton, *Algebraic Curves*, for plane curves and function fields.
- Rick Miranda, *Algebraic Curves and Riemann Surfaces*, for the bridge between compact Riemann surfaces and algebraic curves.
- Joseph H. Silverman and John Tate, *Rational Points on Elliptic Curves*, for elliptic curves in Weierstrass form.

Unclear Source Items

- TODO: unclear from source, page 1. Several arrows in the operation diagram for $(F, +_F, \cdot_F)$ and $(E, +_E, \cdot_E)$ are illegible.
- TODO: unclear from source, page 2. The illustrative quotient/local-chart drawings are not labeled precisely enough to reconstruct as formal diagrams.
- TODO: unclear from source, page 3. Some side annotations near the embedding $M \to \mathbb{P}^N_{\mathbb{C}}$ are not readable.
- TODO: unclear from source, page 6. The explicit torus-to-cubic map is not given.
- TODO: unclear from source, page 8. The square-root sketch does not include branch or ramification details.